

Residential Smart Grid

Report - 03

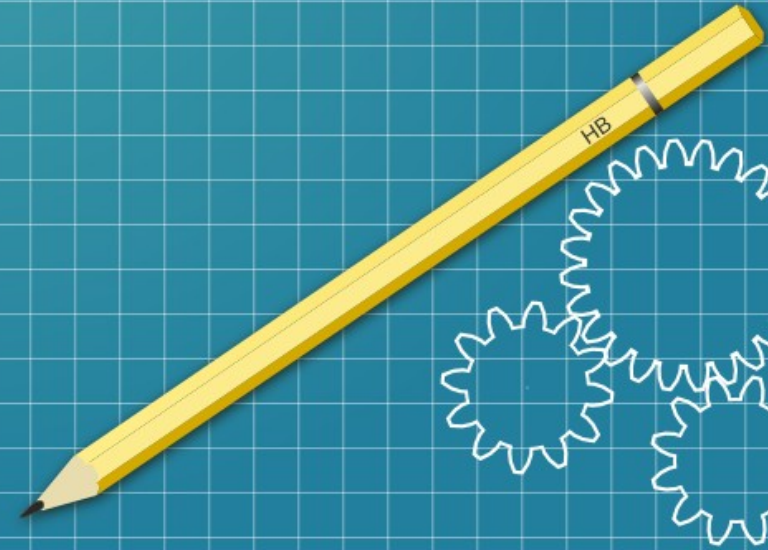
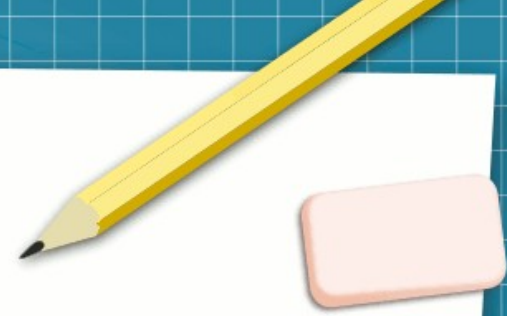


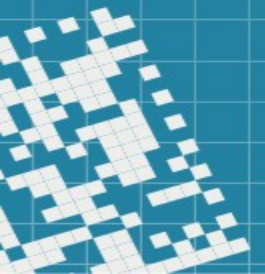
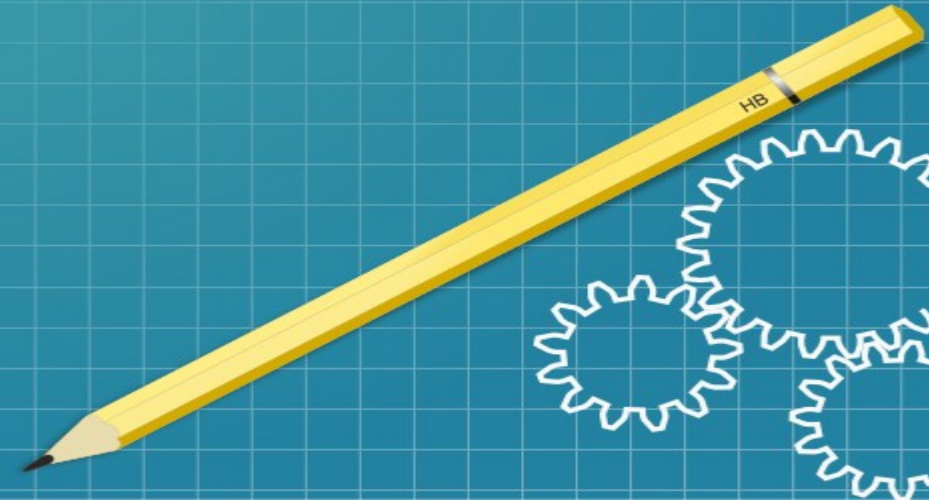
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Project Recap

- Project Description
- Simple System Design
- Detailed System Design

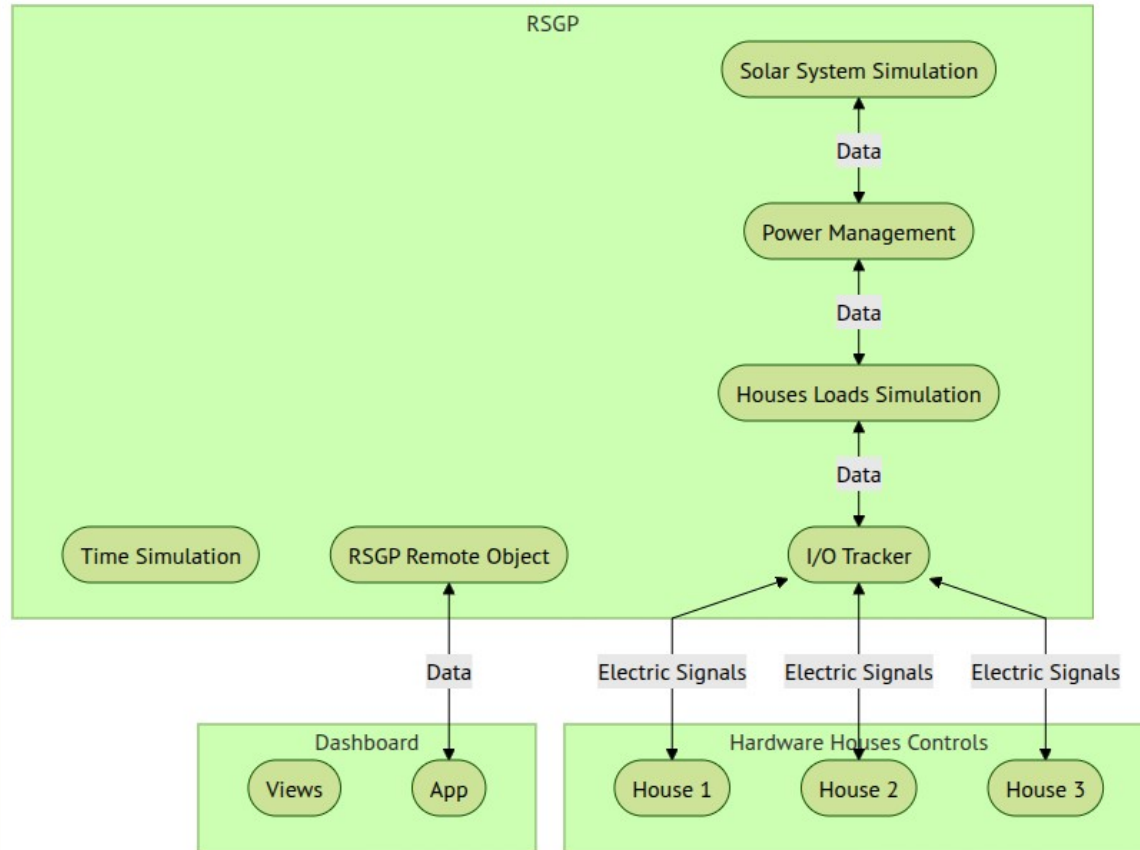


Project Description

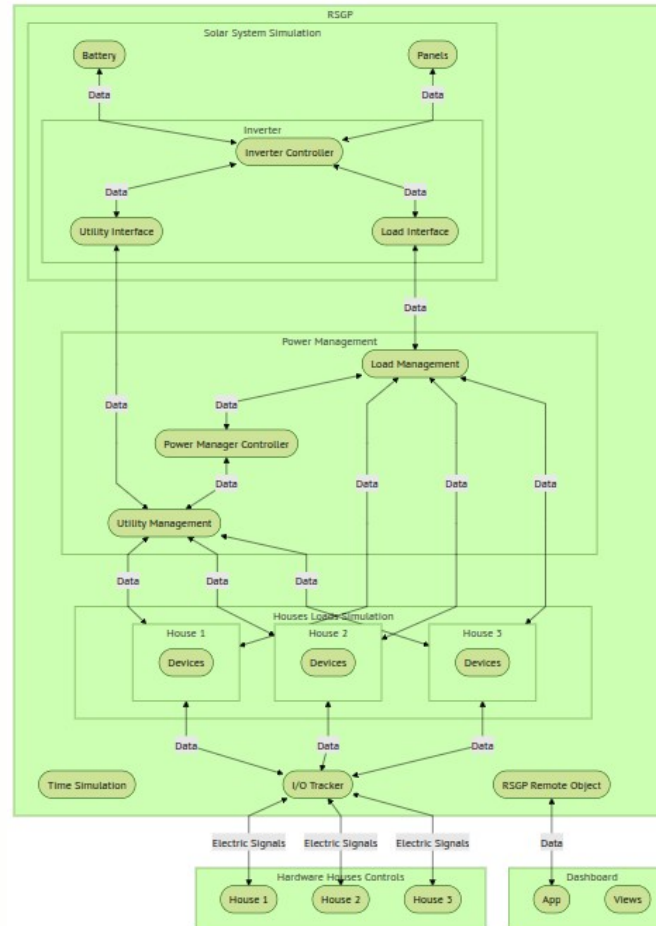


- RSG (Residential Smart Grid) is a grid system that links a large PV system for a residential building with each house in that building. This system controls the power flow between the PV system and each house load and utility lines, using smart techniques and methods. Its aim is to minimize utility dependency and excess PV power.
- Along that grid system is a dashboard to browse and control the system.

Simple System Design



Detailed System Design

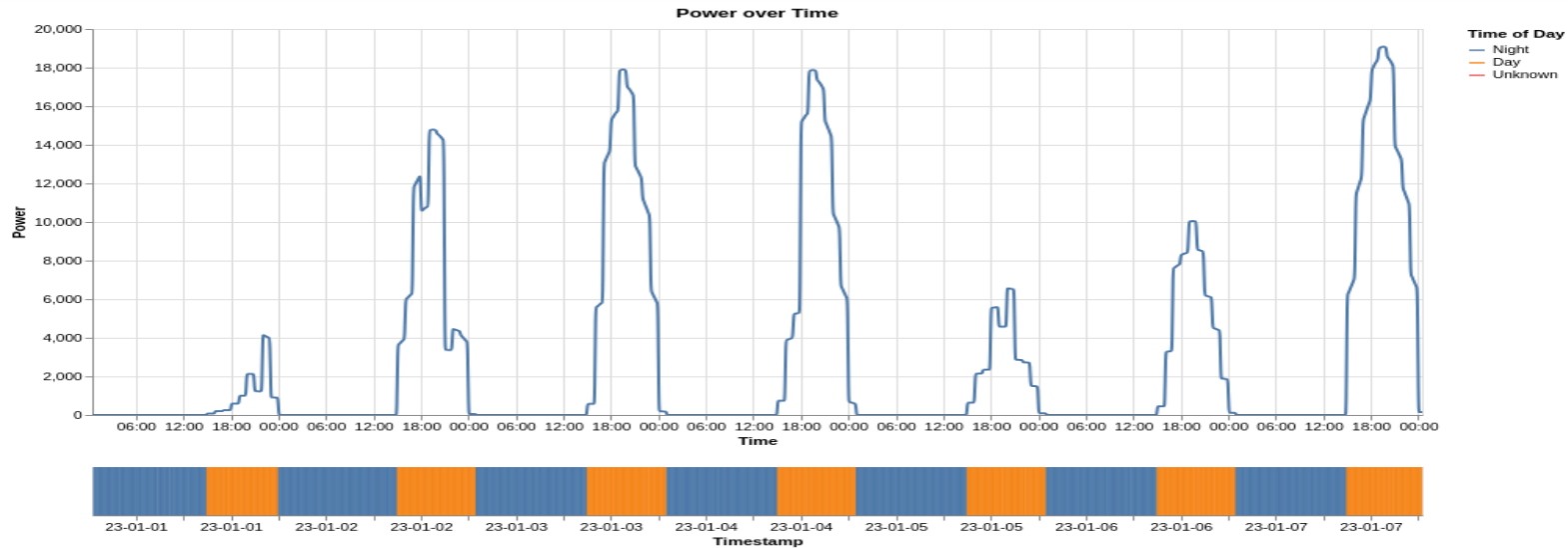


Current Progress & Plans

- RSGP
 - Solar System Simulation
 - Houses Loads Simulation
 - Power Management
 - Remote Object
 - I/O Tracker
- Dashboard
- Notebooks
- Docs

RSGP - Solar System Simulation - Panels

- Uses real data from NSRDB for Arizona. The data includes DNI, DHI, GHI, Zenith Angle, and other related information that is used to calculate the power generated from the panels, using pvlib in Python.

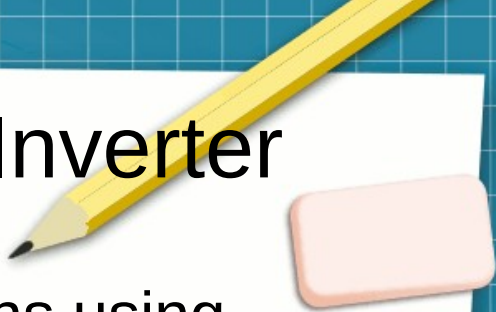


RSGP - Solar System Simulation - Battery

A yellow pencil and a pink eraser are positioned in the top right corner of the slide, appearing to be part of the presentation's design.

- Simulate the battery capacity and battery charging and discharging operations; with maximum charge/discharge power and a specific charging efficiency.

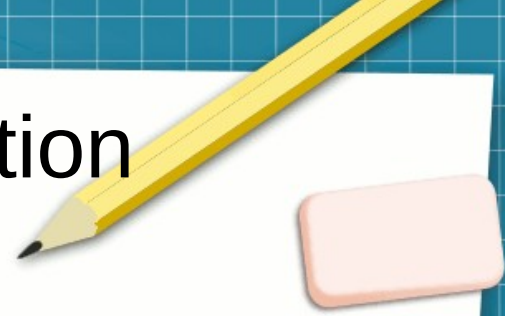
RSGP - Solar System Simulation - Inverter



- Simulate the whole inverter logic and configurations using NREL's PVWatts inverter model via pvlib Python library, including:
 - Converting AC -> DC and DC -> AC
 - Implementing Load Interface and Utility Interface
 - Communicating with Panels, Battery, Load, and Utility with detailed logic for output source priority (i.e. SBU, SUB, USB) and charging priority (i.e. Solar Only, Solar First, Solar + Utility).

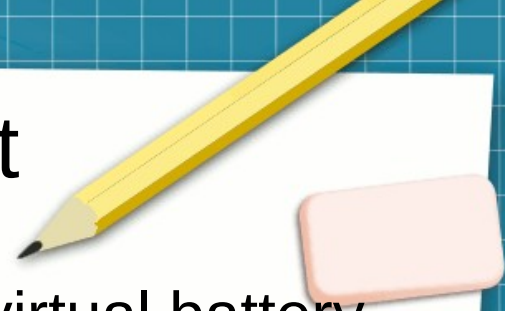
RSGP – Houses Loads Simulation

- Fully implemented. Nothing changed.



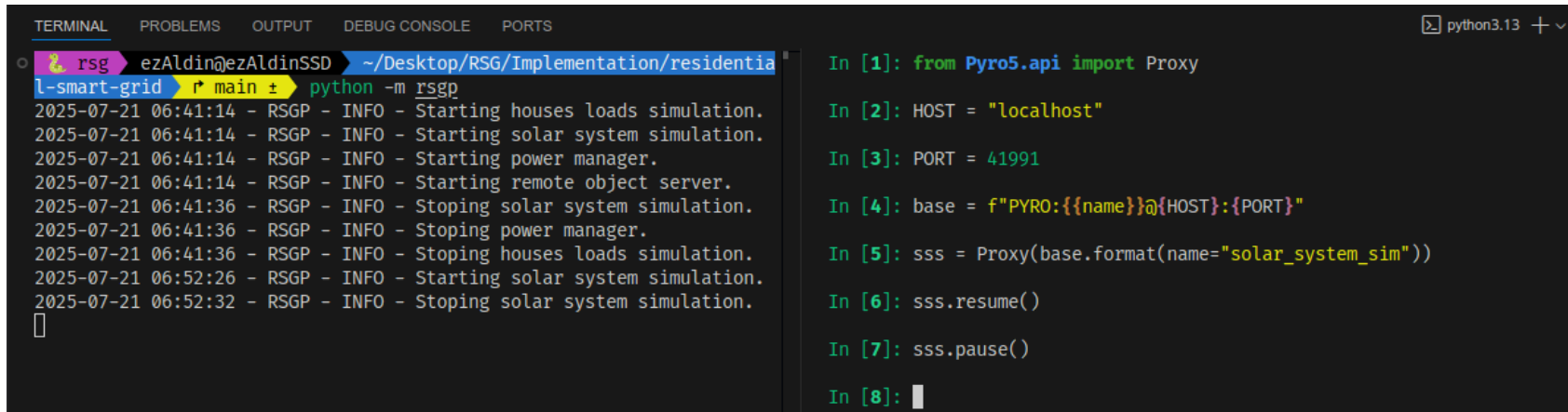
RSGP - Power Management

- We implement a basic solution: Static and equal virtual battery.



RSGP – Remote Object

- We implement a remote object server using Pyro5 Python library.
- That module make it possible to communicate with the running RSGP from outside (other process/program/machine), with a very easy way.



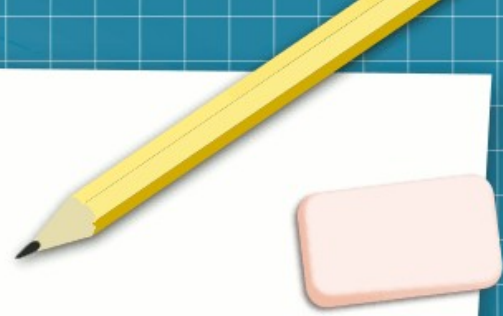
The screenshot shows a code editor with two panels. The left panel is a terminal window showing the execution of a Python script named `rsgp`. The output displays a sequence of log messages indicating the starting and stopping of various simulation components: houses loads simulation, solar system simulation, power manager, and remote object server. The right panel shows an IPython interactive session where the `Pyro5` library is imported, and a `Proxy` object is created to interact with the remote object server.

```
TERMINAL  PROBLEMS  OUTPUT  DEBUG CONSOLE  PORTS  python3.13 + v
```

```
ezAldin@ezAldinSSD ~/Desktop/RSG/Implementation/residential-smart-grid
python -m rsgp
2025-07-21 06:41:14 - RSGP - INFO - Starting houses loads simulation.
2025-07-21 06:41:14 - RSGP - INFO - Starting solar system simulation.
2025-07-21 06:41:14 - RSGP - INFO - Starting power manager.
2025-07-21 06:41:14 - RSGP - INFO - Starting remote object server.
2025-07-21 06:41:36 - RSGP - INFO - Stopping solar system simulation.
2025-07-21 06:41:36 - RSGP - INFO - Stopping power manager.
2025-07-21 06:41:36 - RSGP - INFO - Stopping houses loads simulation.
2025-07-21 06:52:26 - RSGP - INFO - Starting solar system simulation.
2025-07-21 06:52:32 - RSGP - INFO - Stopping solar system simulation.
```

```
In [1]: from Pyro5.api import Proxy
In [2]: HOST = "localhost"
In [3]: PORT = 41991
In [4]: base = f"PYRO:{{name}}@{{HOST}}:{{PORT}}"
In [5]: sss = Proxy(base.format(name="solar_system_sim"))
In [6]: sss.resume()
In [7]: sss.pause()
In [8]:
```

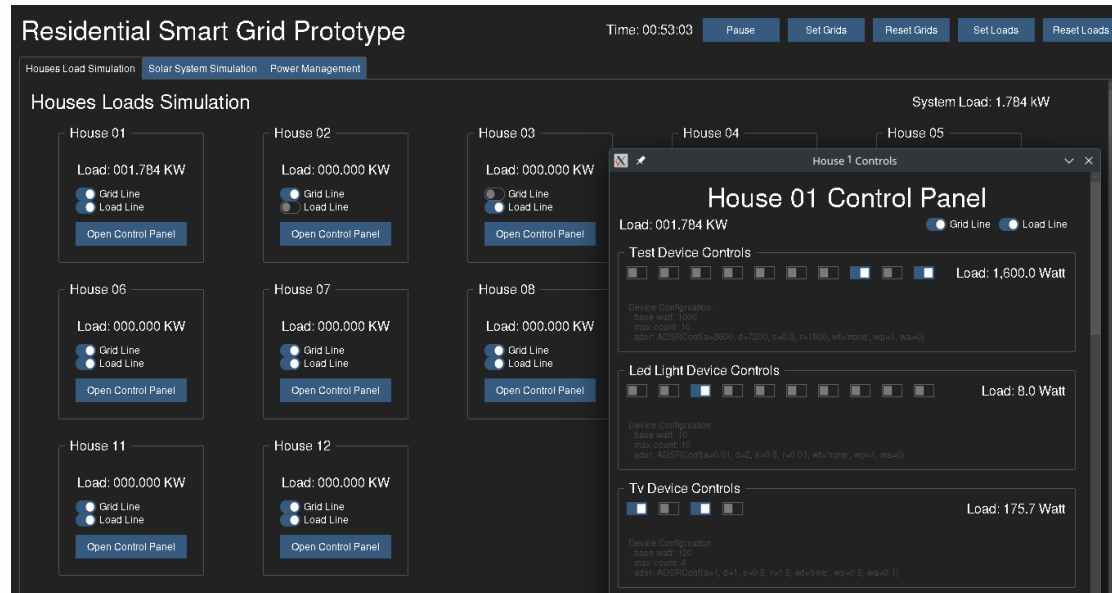
RSGP – I/O Tracker



- Nothing implemented yet. Just ideas.
- It's not a priority, as it's an extra visualization feature, not a core feature.

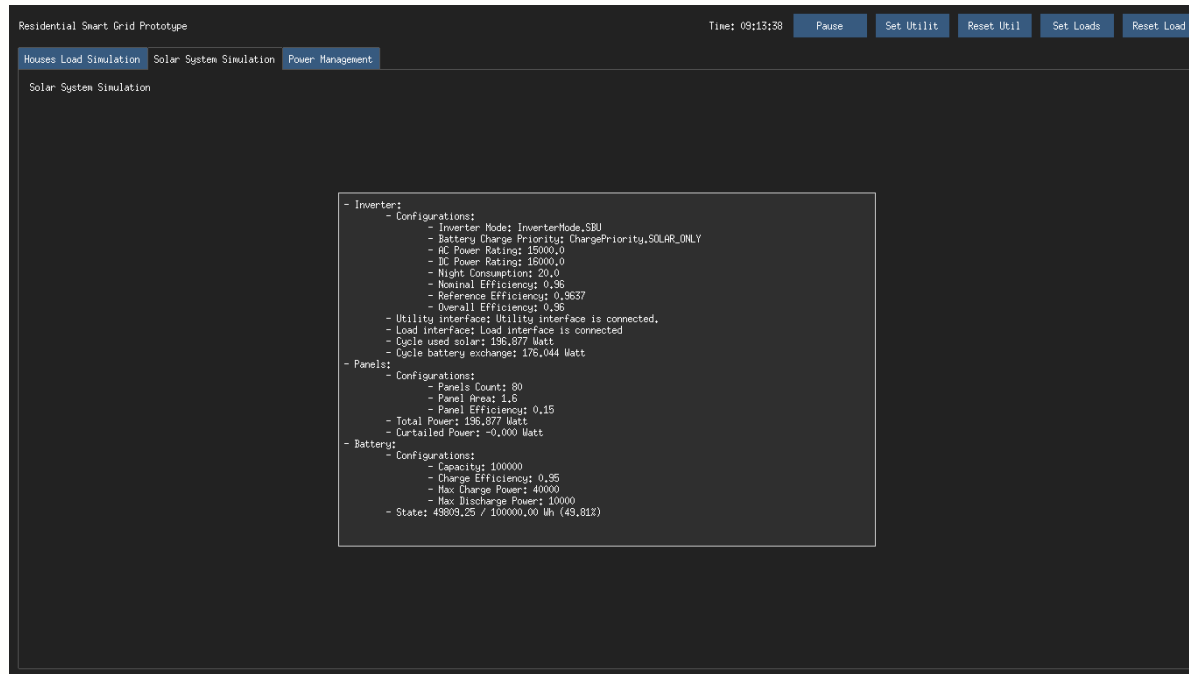
Dashboard

- We decide to make only one dashboard, as prototype, for both admins and householders. The currently dashboard is implemented with tkinter Python library.



Dashboard

- We added some detailed logs for solar system simulation.



```
Residential Smart Grid Prototype
Time: 09:13:38
[Pause] [Set Utilit] [Reset Util] [Set Loads] [Reset Load]

Houses Load Simulation | Solar System Simulation | Power Management

Solar System Simulation

- Inverter:
  - Configurations:
    - Inverter Mode: InverterMode.SBI
    - Battery Charge Priority: ChargePriority.SOLAR_ONLY
    - AC Power Rating: 15000.0
    - DC Power Rating: 15000.0
    - Night Consumption: 20.0
    - Nominal Efficiency: 0.96
    - Reference Efficiency: 0.9637
    - Overall Efficiency: 0.96
  - Utility interface: Utility interface is connected.
  - Load interface: Load interface is connected
  - Cycle used solar: 136.877 Watt
  - Cycle battery exchange: 175.044 Watt
- Panels:
  - Configurations:
    - Panels Count: 80
    - Panel Area: 1.6
    - Panel Efficiency: 0.15
  - Total Power: 136.877 Watt
  - Curtaild Power: -0.000 Watt
- Battery:
  - Configurations:
    - Capacity: 100000
    - Charge Efficiency: 0.95
    - Max Charge Power: 40000
    - Max Discharge Power: 10000
  - State: 43802.25 / 100000.00 Wh (43.81%)
```

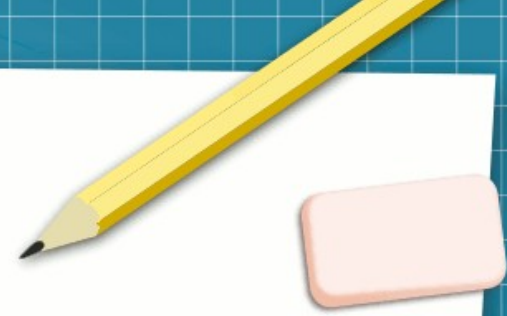
Dashboard



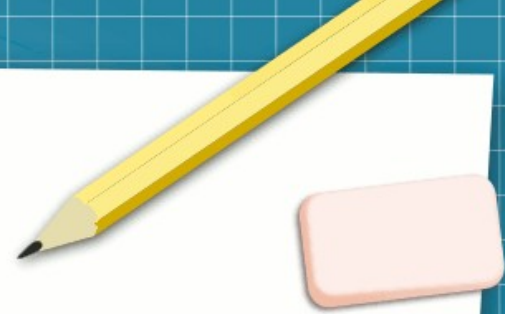
- The dashboard now can run on a separate process, or even on a separate machine, via Remote Object server using Python Pyro5 library – a library for remote object calls in Python.
- We aim to make the dashboard as simple as possible.

Notebooks

- Currently, there exists four notebooks:
 - Mermaid Graphs Editor
 - Inverter Logs Visualization
 - RSGP Logs Visualization
 - NSRDB Visualization



Documentation



- The documentation is partly implemented (~20%)
- The codebase is mostly-documented, so that, the documenting task is not that hard.
- We aim to implement the Documentation in both web-based and PDF theses, via Sphinx.
- We aim to choose a LaTeX template for the theses from some selected templates.